ALS (Amyotrophic Lateral Sclerosis)

Amyotrophic lateral sclerosis (ALS) is a progressive neurodegenerative disease that occurs as a result of the gradual loss of function and death of motor neurons located in the brain and spinal cord and responsible for the control of muscle movements. As motor neurons become damaged, the muscles progressively weaken and undergo atrophy; consequently, voluntary movements such as walking, speaking, swallowing, and breathing become increasingly difficult. The disease follows a progressive course. While sensory functions, bowel and bladder control, cardiac muscle, and generally cognitive functions (intelligence) are not affected, the eye muscles may remain largely preserved until the later stages of the disease or may be almost entirely unaffected.

Worldwide, approximately 2-6 new cases occur per 100,000 people each year. In Europe, the prevalence is around 10-12 per 100,000. It most commonly begins between the ages of 45 and 75, with an average age of onset of 55-65 years. It is slightly more common in men than in women; however, the ratio becomes equal at advanced ages. Approximately 90–95% of cases are sporadic, while 5-10% are familial/genetic. The lifetime risk is approximately 1/350 in men and 1/400 in women.

The initial symptoms are often mild and unilateral: weakness of the thumb and muscle wasting in the hand, foot drop in the leg (tripping while walking), or difficulty with speech or swallowing. Muscle fasciculations, cramps, stiffness, and fatigue are also frequently observed. Patients may mistake these symptoms for “working too hard” or a “pinched nerve.” The symptoms gradually spread to other regions. In approximately one-third of patients, the disease begins in the speech and swallowing muscles; this form may progress more rapidly. Mild cognitive or behavioral changes may occur in approximately half of patients, while frontotemporal dementia accompanies the disease in 10-15% of cases. In advanced stages, walking, speaking, swallowing, and breathing become impaired; patients may become bedridden and may require respiratory support.

The exact cause of this disease is unknown. Genetic factors play a role in approximately 10% of cases; the most common are repeat expansions in the C9orf72 gene and mutations in SOD1, TARDBP, and FUS. In sporadic cases, age, male sex, smoking, high levels of physical activity, head trauma, chemical substances, heavy metals, and certain infections are considered potential risk factors, but a definite association has not been established.

Upper and lower motor neurons in the motor cortex, brainstem, and anterior horn of the spinal cord are lost. In more than 95% of patients, a protein called TDP-43 accumulates within cells. Multiple pathways, including protein degradation, RNA metabolism, the cytoskeleton, and axonal transport, play a role in the disease.

There is no single test for the diagnosis of this disease. The diagnosis is made through neurological examination, EMG (electrical activity of nerves and muscles), and investigations such as blood tests, MRI, and lumbar puncture to exclude conditions that mimic the disease (nerve compression, disc herniation, muscle diseases, and certain cancers). It may be confused with other conditions in the early stages; therefore, careful evaluation is important.

There is no definitive cure for ALS. Riluzole may extend survival by an average of 3-6 months. Edaravone is used in some countries, and tofersen is used in patients with SOD1 mutations. The fundamental approach is multidisciplinary care: physical therapy, speech and swallowing therapy, respiratory support (non-invasive ventilation), and symptom-relieving medications, together with good supportive care and regular follow-up, can significantly affect the quality and duration of life of these patients.

After diagnosis, the average survival time is 2-5 years; however, 5-10% of patients survive for 10 years or longer. Death most often occurs due to respiratory failure. Immobility, inadequate nutrition, and respiratory problems increase the risk of additional complications. Good care and supportive treatment can prolong this period.

Antisense oligonucleotide therapies targeting genetic subtypes and stem cell studies are ongoing. Biomarker research and personalized approaches provide hope, while the importance of multidisciplinary care is increasingly emphasized.

In summary, ALS is a disease characterized by the progressive loss of motor neurons in the brain and spinal cord, leading to muscle weakness and atrophy, and there is currently no definitive cure. It generally occurs in the 50s and is slightly more common in men; symptoms often begin with mild weakness in the hand or leg and gradually spread throughout the body. Its cause is largely unknown, and genetic factors play a role in approximately 10% of cases. Diagnosis is based on clinical findings and EMG, while treatment relies on supportive care. The average survival time is 2-5 years, although this period may be prolonged with good care.

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